

Compact Wideband Multiple-Input- Multiple- Output (MIMO) Antenna design for 5G and IoT Application

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Under the guidance of

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Submitted in partial fulfillment of the requirement of the Degree of Bachelor of Technology
of Maulana Abul Kalam Azad University of Technology, West Bengal

Department of Electronics and Communication Engineering

Vision of the Department

- To build a strong teaching and research environment to cater to the manpower needs in Industrial and Academic domains of the rapidly growing Electronics and Communication Engineering.

Mission of the Department

- To produce certified industry-ready professional in Electronics and Communication Engineering, through innovative educational programs incorporating laboratory practices and project-based teaching-learning processes, in a modern environment.
- To create knowledge base of advanced technologies through research in the area of Electronics and Communication, for competitive and sustainable development of the country.
- To groom the department as a learning center to inculcate advancement of technology in Electronics and Communication Engineering with social values and environmental awareness.

Program Specific Outcome (PSOs)

After completion of program graduate engineer would have:

- **PSO1. Professional skills:** An ability to apply the knowledge in Electronics and Communication Engineering in various areas, like Communications, Signal processing, VLSI and Embedded Systems.
- **PSO2. Competency:** An ability to qualify at the State, National and International level competitive examinations for employment, higher studies and research

Program Outcome

Engineering Graduates will be able to:

- **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- **Conduct investigations of complex problems:** Use research-based knowledge and research methods

effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Course Outcomes for Final Year Project

After completion of the project, student will be able to:

CO no.	Outcome Statements	Bloom's Level
CO1	Apply technical knowledge in the solution of complex real-life problems related to public health and safety, culture, society, and environment	3
CO2	Review research literature and use the research-based knowledge to identify, formulate, and analyze the problem	5
CO3	Design innovative solutions for complex engineering problems, which will be published as research paper or developed as a marketable product in future	6
CO4	Apply modern tools to design and develop the solution to problem	3
CO5	Assess societal, health, safety, legal and cultural issues relevant to project and the responsibilities relevant to the professional engineering practice.	5
CO6	Evaluate the impact on the environment of working project	5
CO7	Apply ethical principles and commit to ethics and responsibilities related to engineering practice.	3
CO8	Function effectively as a member or leader of a team	3
CO9	Communicate effectively on professional activities with the team members, superiors and with society at large	3
C10	Plan, manage the project and control finance as a member and leader in a team.	3
C11	Apply the knowledge acquired during the project, in future higher studies or professional job.	4

Bloom's Level: Remember = 1, Understand = 2, Apply = 3, Analyze = 4, Evaluate = 5, Create = 6

PO 1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization for the solution of complex engineering problems.	
Competency	Indicators
1.1 Demonstrate competence in mathematical modelling	1.1.1 Apply mathematical techniques such as calculus, linear algebra, and statistics to solve problems 1.1.2 Apply advanced mathematical techniques to model and solve Electronics engineering problems
1.2 Demonstrate competence in basic sciences	1.2.1 Apply laws of natural science to an engineering problem
1.3 Demonstrate competence in engineering fundamentals	1.3.1 Apply fundamental engineering concepts to solve engineering problems
1.4 Demonstrate competence in specialized engineering knowledge to	1.4.1 Apply Electronics engineering concepts to solve engineering problems.

the program

Program Outcomes -Competencies – Performance Indicators (PIs)

PO 2: Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

Competency	Indicators
2.1 Demonstrate an ability to identify and formulate complex engineering problem	2.1.1 Articulate problem statements and identify objectives 2.1.2 Identify engineering systems, variables, and parameters to solve the problems 2.1.3 Identify the mathematical, engineering and other relevant knowledge that applies to a given problem
2.2 Demonstrate an ability to formulate a solution plan and methodology for an engineering problem	2.2.1 Reframe complex problems into interconnected sub-problems 2.2.2 Identify, assemble and evaluate information and resources. 2.2.3 Identify existing processes/solution methods for solving the problem, including forming justified approximations and assumptions 2.2.4 Compare and contrast alternative solution processes to select the best process.
2.3 Demonstrate an ability to formulate and interpret a model	2.3.1 Combine scientific principles and engineering concepts to formulate model/s (mathematical or otherwise) of a system or process that is appropriate in terms of applicability and required accuracy. 2.3.2 Identify assumptions (mathematical and physical) necessary to allow modelling of a system at the level of accuracy required
2.4 Demonstrate an ability to execute a solution process and analyze results	2.4.1 Apply engineering mathematics and computations to solve mathematical models 2.4.2 Produce and validate results through skillful use of contemporary engineering tools and models 2.4.3 Identify sources of error in the solution process, and limitations of the solution. 2.4.4 Extract desired understanding and conclusions consistent with objectives and limitations of the analysis

PO 3: Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations. .

Competency	Indicators
3.1 Demonstrate an ability to define a complex/ open-ended problem in engineering terms	3.1.1 Recognize that need analysis is key to good problem definition 3.1.2 Elicit and document, engineering requirements from stakeholders 3.1.3 Synthesize engineering requirements from a review of the state-of-the-art 3.1.4 Extract engineering requirements from relevant engineering Codes and Standards such as ASME, ASTM, BIS, ISO and ASHRAE. 3.1.5 Explore and synthesize engineering requirements considering health, safety risks, environmental, cultural and societal issues 3.1.6 Determine design objectives, functional requirements and arrive at specifications
3.2 Demonstrate an ability to generate a diverse set of alternative design solutions	3.2.1 Apply formal idea generation tools to develop multiple engineering design solutions 3.2.2 Build models/prototypes to develop a diverse set of design solutions 3.2.3 Identify suitable criteria for the evaluation of alternate design solutions.
3.3 Demonstrate an ability to select an optimal design scheme for further development	3.3.1 Apply formal decision-making tools to select optimal engineering design solutions for further development 3.3.2 Consult with domain experts and stakeholders to select candidate engineering design solution for further development

3.4 Demonstrate an ability to advance an engineering design to defined end state	3.4.1 Refine a conceptual design into a detailed design within the existing constraints (of the resources) 3.4.2 Generate information through appropriate tests to improve or revise the design
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PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

Competency	Indicators
4.1 Demonstrate an ability to conduct investigations of technical issues consistent with their level of knowledge and understanding	4.1.1 Define a problem, its scope and importance for purposes of investigation 4.1.2 Examine the relevant methods, tools and techniques of experiment design, system calibration, data acquisition, analysis and presentation 4.1.3 Apply appropriate instrumentation and/or software tools to make measurements of physical quantities 4.1.4 Establish a relationship between measured data and underlying physical principles
4.2 Demonstrate an ability to design experiments to solve open-ended problems	4.2.1 Design and develop an experimental approach, specify appropriate equipment and procedures 4.2.2 Understand the importance of the statistical design of experiments and choose an appropriate experimental design plan based on the study objectives
4.3 Demonstrate an ability to analyze data and reach a valid conclusion	4.3.1 Use appropriate procedures, tools and techniques to conduct experiments and collect data 4.3.2 Analyze data for trends and correlations, stating possible errors and limitations 4.3.3 Represent data (in tabular and/or graphical forms) so as to facilitate analysis and explanation of the data, and drawing of conclusions 4.3.4 Synthesize information and knowledge about the problem from the raw data to reach appropriate conclusions

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

Competency	Indicators
5.1 Demonstrate an ability to identify / create modern engineering tools, techniques and resources	5.1.1 Identify modern engineering tools such as computer-aided drafting, modeling and analysis; techniques and resources for engineering activities 5.1.2 Create/adapt/modify/extend tools and techniques to solve engineering problems
5.2 Demonstrate an ability to select and apply discipline-specific tools, techniques and resources	5.2.1 Identify the strengths and limitations of tools for (i) acquiring information, (ii) modeling and simulating, (iii) monitoring system performance, and (iv) creating engineering designs. 5.2.2 Demonstrate proficiency in using discipline-specific tools
5.3 Demonstrate an ability to evaluate the suitability and limitations of tools used to solve an engineering problem	5.3.1 Discuss limitations and validate tools, techniques and resources 5.3.2 Verify the credibility of results from tool use with reference to the accuracy and limitations, and the assumptions inherent in their use.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

Competency	Indicators
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6.1 Demonstrate an ability to describe engineering roles in a broader context, e.g. pertaining to the environment, health, safety, legal and public welfare	6.1.1 Identify and describe various engineering roles; particularly as pertains to protection of the public and public interest at the global, regional and local level
6.2 Demonstrate an understanding of professional engineering regulations, legislation and standards	6.2.1 Interpret legislation, regulations, codes, and standards relevant to your discipline and explain its contribution to the protection of the public

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and the need for sustainable development.

Competency	Indicators
7.1 Demonstrate an understanding of the impact of engineering and industrial practices on social, environmental and in economic contexts	7.1.1 Identify risks/impacts in the life-cycle of an engineering product or activity 7.1.2 Understand the relationship between the technical, socio-economic and environmental dimensions of sustainability
7.2 Demonstrate an ability to apply principles of sustainable design and development	7.2.1 Describe management techniques for sustainable development 7.2.2 Apply principles of preventive engineering and sustainable development to an engineering activity or product relevant to the discipline

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice

Competency	Indicators
8.1 Demonstrate an ability to recognize ethical dilemmas	8.1.1 Identify situations of unethical professional conduct and propose ethical alternatives
8.2 Demonstrate an ability to apply the Code of Ethics	8.2.1 Identify tenets of the ASME professional code of ethics 8.2.2 Examine and apply moral & ethical principles to known case studies

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Competency	Indicators
9.1 Demonstrate an ability to form a team and define a role for each member	9.1.1 Recognize a variety of working and learning preferences; appreciate the value of diversity on a team 9.1.2 Implement the norms of practice (e.g. rules, roles, charters, agendas, etc.) of effective team work, to accomplish a goal.
9.2 Demonstrate effective individual and team operations-- communication, problem-solving, conflict resolution and leadership skills	9.2.1 Demonstrate effective communication, problem-solving, conflict resolution and leadership skills 9.2.2 Treat other team members respectfully 9.2.3 Listen to other members 9.2.4 Maintain composure in difficult situations
9.3 Demonstrate success in a team-based project	9.3.1 Present results as a team, with smooth integration of contributions from all individual efforts

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions

Competency	Indicators
10.1 Demonstrate an ability to comprehend technical literature and document project work	10.1.1 Read, understand and interpret technical and non-technical information 10.1.2 Produce clear, well-constructed, and well-supported written engineering documents

	10.1.3 Create flow in a document or presentation - a logical progression of ideas so that the main point is clear
10.2 Demonstrate competence in listening, speaking, and presentation	10.2.1 Listen to and comprehend information, instructions, and viewpoints of others 10.2.2 Deliver effective oral presentations to technical and non-technical audiences
10.3 Demonstrate the ability to integrate different modes of communication	10.3.1 Create engineering-standard figures, reports and drawings to complement writing and presentations 10.3.2 Use a variety of media effectively to convey a message in a document or a presentation

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's work, as a member and leader in a team, to manage projects and in multidisciplinary environments. .

Competency	Indicators
11.1 Demonstrate an ability to evaluate the economic and financial performance of an engineering activity	11.1.1 Describe various economic and financial costs/benefits of an engineering activity 11.1.2 Analyze different forms of financial statements to evaluate the financial status of an engineering project
11.2 Demonstrate an ability to compare and contrast the costs/benefits of alternate proposals for an engineering activity	11.2.1 Analyze and select the most appropriate proposal based on economic and financial considerations.
11.3 Demonstrate an ability to plan/manage an engineering activity within time and budget constraints	11.3.1 Identify the tasks required to complete an engineering activity, and the resources required to complete the tasks. 11.3.2 Use project management tools to schedule an engineering project, so it is completed on time and on budget.

PO 12: Life-long learning: Recognise the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change..

Competency	Indicators
12.1 Demonstrate an ability to identify gaps in knowledge and a strategy to close these gaps	12.1.1 Describe the rationale for the requirement for continuing professional development 12.1.2 Identify deficiencies or gaps in knowledge and demonstrate an ability to source information to close this gap
12.2 Demonstrate an ability to identify changing trends in engineering knowledge and practice	12.2.1 Identify historic points of technological advance in engineering that required practitioners to seek education in order to stay current 12.2.2 Recognize the need and be able to clearly explain why it is vitally important to keep current regarding new developments in your field
12.3 Demonstrate an ability to identify and access sources for new information	12.3.1 Source and comprehend technical literature and other credible sources of information 12.3.2 Analyze sourced technical and popular information for feasibility, viability, sustainability, etc.

PSO1: Professional skills: An ability to apply the knowledge in Electronics and Communication Engineering in various areas, like Communications, Signal processing, VLSI and Embedded Systems. ..
PSO2: Competency: An ability to qualify at the State, National and International level competitive examinations for employment, higher studies and research

Competency	Indicators
PSO1.1 PSO2.1 Demonstrate an ability to apply knowledge in Communications, Signal Processing, VLSI and Embedded Systems Qualify top level competitive examinations for employment, higher studies and research	PSO1.1.1 Design circuitry and systems related to Communications, Signal Processing, VLSI and Embedded Systems PSO2.1.1 Qualify top level competitive examinations for employment PSO2.1.2 Qualify top level competitive examinations for higher studies and research PSO1.1.2 Provide solution to the problems related to the specified systems PSO1.1.3 Upgrade the systems with latest technology and incorporating more facilities

Rubrics for Percentage of Performance Indicator and Mapping Grade in CO-PO matrix

Mapping ratio	Strength
>66%	3
<66% but >= 33%	2
<33% but >= 0%	1

CO-PO

Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2			2	1				3		1	1	2	2
CO2	2	2	3	2	1	1			2			1	2	
CO3	3		3	1		2	2	2	2				3	
CO4	3		2				1						2	3
CO5	3		2			1	2	1						
CO6	2		2		1		3		1				2	3
CO7		2		3	1		3		1				1	
CO8			3		3	2		1		3			2	
CO9	2		3		1				2					
CO10				2		3	2		1				1	2
CO11	3		2		1	3		1					3	2

Program level Course-PO matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO	1.82	0.36	1.82	0.91	0.82	1.09	1.18	0.45	1.09	0.27	0.09	0.18	1.64	1.09

Project Planning & Finance:

Implementation of the project is planned and documented. Implementation plan is broken into tasks, duration and owner of the tasks. The **Gantt Chart** of the project is drawn and included in the report.

Cost of the components and any other expenses related to the project is accounted and mentioned in the report.

Acknowledgement

We would like to thank our project guide Dr. Swarup Das who gave us this opportunity to work on this project. We got to learn a lot from this project about designing of Wideband MIMO Antenna for 5G and IoT application. We would also like to thank our HoD Dr. Prasun Chowdhury and other faculty members of ECE Department.

At last, we would like to extend my heartfelt thanks to our parents because without their help this project would not have been successful. Finally, I would like to thank my dear friends who have been with me all the time.

Sayantana Mandal (University Roll No.:12200319036),
Aritra Pal (University Roll No.:12200319024),
Maitreya Gupta (University Roll No.:12200319005),
Debajyoti Kansabanik (University Roll No.:12200319038)

Certificate

Department of Electronics and Communication Engineering

St. Thomas' College of Engineering and Technology

This is to certify that the project entitled “**Compact Wideband Multiple-Input-Multiple- Output (MIMO) Antenna design for 5G and IoT Application**”, has been carried out by

Sayantana Mandal (University Roll No.:12200319036),
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Contents

Chapter 1

Introduction

1.1 Introduction

The use of wireless devices is the latest trend in communication technology, and there is a constant demand for compactness or miniaturization of wireless electronic devices, as well as an increase in speed and data rate for these devices. Antenna systems were developed to transport signals at a speedy rate with a lot higher capacity, thanks to antennas with a broad range and high gain as well as the quick development of communication technology. Ultrawideband(UWB) technology has rapidly gained popularity and demand for recent wireless communication systems after the allocation of 3.1-10.6 GHz by the Federal Communications Commission (FCC) for UWB applications. Since then, a myriad of research opportunities and challenges exist for the design of Wide Band antenna systems for application in high speed wireless devices. Multiple-Input-Multiple-Output (MIMO) systems provide a significant increase in channel capacity without the need of additional bandwidth or transmit power by deploying multiple antennas for transmission to achieve an array gain and diversity gain, thereby improving the spectral efficiency and reliability. Since MIMO systems employ multiple antennas, they require high decoupling between antenna elements. Overall WB MIMO systems require a high isolation of less than -16 dB and also a compact size for compatibility with integrated circuits.

1.2 Project Overview

This thesis focuses on the analysis and design of MIMO antennas with a compact planar profile that have an operating range in the entire UWB (3.1- 10.6 GHz) and desired antenna performance characteristics. Recently, academic and industrial communities have realized the trade-offs between antenna design and transceiver complexity. In general, the transceiver complexity has been increased, with the introduction of advanced wireless transmission techniques. In order to enhance the performance of transceiver without sacrificing its costly architecture, advanced antenna design should be used as the antenna is an integral part of the transceiver. Also, the complexity of the overall transceiver is reduced.[1]

This dissertation presents the work on the design of four-element MIMO antennas and various isolation structures and mechanisms to reduce the mutual coupling between the two elements, out of which two major antenna designs are proposed and analyzed separately for their isolation, bandwidth and radiation characteristics. First, a printed Wideband (WB) MIMO antenna system is proposed for portable WB applications. This MIMO antenna system consists of four octagonal radiating elements on a single low-cost FR4($\epsilon=4.3$) substrate of a compact size of 50 mm $\times$ 50 mm and is fed by a 50 Ω microstrip line. The proposed antenna system operates over a wide frequency range from 2.64 GHz to 20 GHz . Simulated results of S-parameters of the proposed antenna system are obtained and a high isolation of greater than -20 dB is achieved in the sub-6 GHz 5G band, which is found suitable for MIMO applications. To implement WB technology, there are many challenges to overcome. WB has a significant effect on antenna design. It has attracted a surge of interest in antenna design by providing new challenges and opportunities for antenna designers as WB systems require an antenna with an operating bandwidth covering the entire WB (3.1-10.6 GHz) and capable of receiving on associated frequencies at the same time [2]. Consequently, the antenna behavior and performance have to be consistent and predictable across the WB. Moreover, WB is a technology

that modulates impulse based waveforms rather than continuous carrier waves. Hence, the design of WB antennas requires different considerations from those used in designing narrowband antennas. The hardest challenge in designing a WB antenna is to attain wide impedance bandwidth with high radiation efficiency.

The concurrent surge of wireless devices, with high level of miniaturization and high frequency of operation, has enhanced the interest in designing high performance antenna types. Therefore, there is a growing demand for small and low cost WB antennas that are able to provide satisfactory performance in both time and frequency domains. The trend in recent wireless systems, including WB based systems, are to build small, low-profile integrated circuits so as to be compatible with portable wireless devices. Also, the size affects the gain and bandwidth. Therefore, the size of the antenna is considered as one of the critical issues in WB system design. The use of a planar design can minimize the volume of the WB antennas by replacing three-dimensional radiators with their planar versions. Also, the two-dimensional (2D) geometry makes the fabrication easier. As a result, the planar antenna can be printed on a PCB and thus can be easily integrated into RF circuits [3].

Recently, there is a demand to increase the data rate of existing wireless communication systems. The application of diversity techniques, most commonly assuming two antennas in a mobile terminal, can enhance the data rate and reliability without sacrificing additional spectrum or transmitted power in rich scattering environments. MULTIPLE-INPUTMULTIPLE-OUTPUT (MIMO) technology has attracted attention in modern wireless communication systems Multiple-input-multiple-output (MIMO) systems transmit the same power using multiple antennas at the transmitter and receiver thereby increasing the channel capacity without the need of additional bandwidth or power. MIMO WB systems can further increase the channel capacity as compared to conventional MIMO systems for narrowband applications. To combat the multipath fading problem in an indoor UWB wireless communication system, an WB diversity antenna system is a promising candidate. However, for an efficient MIMO antenna system mutual coupling between the individual antennas should be as low as possible.

Hence, these design challenges and features for achieving high channel capacity with less complexity kindles the interest and serves as a motivation to the researchers in the study and design of MIMO antennas for high data rate UWB applications.

1.3 Project Objective

This project presents the design of a wideband octagonal microstrip patch MIMO antenna for 5G and IoT application which is to reduce the mutual coupling between the four elements, out of which two major antenna designs are proposed and analyzed separately for their isolation, bandwidth and radiation characteristics. A four-element MIMO antenna system has been built employing orthogonal polarization diversity. The patch antennas are placed on the top of 1.6 mm thick FR-4 substrate having a length of 50 mm, a width of 50 mm, and relative permittivity of 4.3 ($\epsilon=4.3$). The proposed MIMO antenna operates throughout 2.64 GHz-20GHz, which can be used for IoT and Sub-6GHz 5G services all over the world. The results of the designed structure have been simulated in a finite element method-based solver Computer Simulation Technique (CST) *ie* CST Studio Suite. A four-element MIMO antenna system has been built employing orthogonal polarization diversity.

1.4 Application

Microstrip antennas were initially used in military systems and satellites. Recently, these antennas are being used for commercial applications due to reduced cost. Some of the applications are listed below:

- Environmental instrumentation and remote sensing.
- Biomedical instruments.
- Mobile handsets.
- Smart devices *eg.* Smart watch, Smart fan, Alexa etc.
- Wireless devices *eg.* Wireless mouse, USB dongle etc.
- Bluetooth and Wi-Fi devices.
- Commercial aircraft and missiles.
- Radio Frequency(RF) Biosensors.

1.5 Project Problem Statement

The Bandwidth, gain, isolation of MIMO Antenna are very less and the size of it is very large. To design and optimize a MIMO antenna to miniaturize it and to increase the gain, bandwidth, isolation of the MIMO antenna for 5G and IoT application.

Chapter 2

Background of the Project

2.1 Literature Survey

2.1.1 “A Compact Koch Fractal UWB MIMO Antenna with WLAN Band-Rejection” – S. Tripathi et.al.

In that paper, a compact octagonal shaped fractal ultrawideband multiple-input-multiple-output (MIMO) antenna is presented and its characteristics are investigated. In order to achieve the desired miniaturization and wideband phenomena, self-similar and space filling properties of Koch fractal geometry are used in the antenna design. These fractal monopoles are placed orthogonal to each other for good isolation. Moreover, grounded stubs are used in the geometry to provide further improvement in the isolation. The band rejection phenomenon in wireless local area network (WLAN) band is achieved by etching a C-shaped slot from the monopole of the antenna. The proposed antenna has compact dimensions of 45 mm × 45 mm and exhibits quasi-omnidirectional radiation pattern. In addition, it shows an impedance bandwidth ($S_{11} < -10\text{dB}$) from 2-10.6 GHz with isolation better than 17dB over the entire UWB range. Diversity performance is also evaluated in terms of ECC and capacity loss. The measured results show good agreement with the simulated ones.[4]

2.1.2 “Compact Dual Polarized UWB Quasi-Self-Complementary MIMO/Diversity Antenna with Band-Rejection Capability” - Jianfeng Zhu et.al.

A novel compact UWB multiple-input-multiple-output (MIMO) antenna system with dual polarization and band-rejection capabilities is proposed. The proposed MIMO antenna system consists of two quasi-self-complementary (QSC) antenna elements. The elements are arranged orthogonally and fed perpendicularly to obtain polarization diversity. High isolation can be achieved without additional decoupling structure owing to the inherent advantage of the self-complementary structure. Notched band at WLAN system can be realized by etching a bent slit in each of the radiating elements. Besides, a four-element MIMO system is also proposed and investigated to fully reveal its potential use. Diversity performance in terms of envelop correlation coefficient (ECC) and the mean effective gain (MEG) ratio are studied. Measured results show that the proposed antenna has a wide bandwidth ranging from 3 to 12 GHz with band rejection at WLAN system and high port isolation ($S_{12} \leq -20\text{ dB}$ at most of the band), which demonstrate the proposed MIMO/diversity antenna system can be a good candidate for UWB applications.[5]

2.1.3 “Planar UWB MIMO Antenna with Pattern Diversity and Isolation Improvement for Mobile Platform Based on the Theory of Characteristic Modes” - Xing Zhao et.al.

This paper presents a planar ultra-wideband multi-input multi-output antenna with pattern diversity and isolation improvement based on the theory of characteristic modes. The proposed antenna (for new-generation wireless mobile terminals) consists of two ultra-wideband antenna elements — a compact quarter loop and a circular planar

monopole located on the opposite edges of a rectangular ground plane. These two antenna elements are able to excite different modes in the ground plane and thus realize the desired diverse radiation patterns and high isolation without any additional decoupling structures. The experimental results show that both antenna elements are ultra-wideband with impedance bandwidths (return losses more than 6 dB for the quarter-loop antenna and more than 10 dB for the circular planar monopole antenna) of 130% (2–9.5 GHz). The isolation is over 20 dB with a peak value of 50 dB.

The realized gains are above 1.5 dBi. The total efficiencies are above 70%. The envelope correlation coefficient is less than 0.03 assuming that the antenna operates in an isotropic channel.[6]

2.1.4 “Design of a Planar MIMO Antenna for LTE-Advanced” -A.A. Asaker et. al.

In this paper presents the design of a wideband microstrip patch antenna for LTE-A. The bandwidth of the conventional patch is enlarged by using etched slots at the antenna patch. The designed antenna has been fabricated by using thin film and photolithographic technique and has been measured by using the Vector Network Analyzer. The simulated and measured results were found to have good match with each other. Then, by using the designed single element antenna, a four-element MIMO antenna system has been built employing orthogonal polarization diversity. Isolation between the microstrip elements is increased by placing metal structure between antenna elements. For more isolation between antenna elements, Slotted Ground Plane SGP is utilized. It is found, by using commercial software CST Microwave Studio and measurement, that the designed planar MIMO antenna system has sufficiently high return loss and low mutual coupling at the required bandwidth of 70 MHz. It is found also that the developed antenna system meets the requirements for LTE-Advanced (2500 – 2570 MHz) band “CA-B7” as of today’s standard based on 36.101 Table 5.5-1 (March 2012) .[7]

2.1.5 “Mutual Coupling Reduction for UWB MIMO Antennas with a Wideband Neutralization Line” - Shuai Zhang et.al.

A wideband neutralization line is proposed to reduce the mutual coupling of a compact ultrawideband (UWB) MIMO

antenna. With the introduced decoupling method, the designed UWB MIMO antenna covers the band of 3.1-5 GHz with an isolation of higher than 22 dB. The proposed wideband neutralization line is not necessarily placed in the clearance area between two MIMO elements and can be put above the copper ground. A small clearance (antenna area) of 35 mm × 16 mm is achieved. The designed UWB MIMO antenna is fabricated. S parameters, radiation patterns, total efficiency and realized gain of the prototype are measured and compared with the simulations.[8]

2.1.6 “Closely-Packed UWB MIMO/Diversity Antenna With Different Patterns and Polarizations for USB Dongle Applications” - Shuai Zhang et.al.

A closely-packed ultrawideband (UWB) multiple-input multiple-output (MIMO)/diversity antenna (of two elements) with a size of 25 mm by 40 mm is proposed for USB dongle applications. Wideband isolation can be achieved through the different patterns and polarizations of the two antenna elements. Moreover, the slot that is formed between the monopole and the ground plane of the half slot antenna is conveniently used to further enhance the isolation at the lower frequencies and to provide an additional resonance at one antenna element in order to

increase its bandwidth. The underlying mechanisms of the antenna's wide impedance bandwidth and low mutual coupling are analyzed in detail. Based on the measurement results, the proposed antenna can cover the lower UWB band of 3.1–5.15 GHz, and within the required band, the isolation exceeds 26 dB. The gains and total efficiencies of the two antenna elements are also measured. Furthermore, a chassis mode can be excited when a physical connection is required between the ground planes of the two antenna elements. Without affecting the performance of the half slot element, the monopole can now cover the band of 1.78–3 GHz, apart from the UWB band. The proposed

antenna structure is found to provide good MIMO/diversity performance, with very low envelope correlation of less than 0.1 across the UWB band.[9]

2.1.7 “An Ultrawideband MIMO Antenna with Dual Band-Notched Function” - Tzu-Chun Tang et.al.

A new technique to achieve both good isolation and dual band-notched functions for ultrawideband (UWB) multiple-input-multiple-output (MIMO) antenna was developed. The proposed antenna consists of the radiation patch connected through a via with the strip placed beneath the patch. The strip not only provides another coupling path but also serves as the impedance transformer, resulting in good isolation ($S_{21} < -15$ dB) and dual band rejection at WiMAX (3.4–3.7 GHz) and WLAN (5.15–5.35 and 5.725–5.825 GHz) over the UWB system operation (3.1–10.6 GHz). Without decoupling element or resonator which is often placed between the antennas to acquire good isolation or band rejection, the proposed UWB MIMO antenna requires an individual antenna element size of only 11.2×12 mm² ($0.12\lambda_0 \times 0.11\lambda_0$) and overall MIMO antenna size of 30×40 mm² ($0.31\lambda_0 \times 0.41\lambda_0$). The edge to edge separation between antennas is $0.052\lambda_0$ and separation between feeds is $0.175\lambda_0$ at 3.1 GHz. Detailed investigations of the parameters were performed and the experimental results agree closely with the simulation results, demonstrating the feasibility of the proposed method.[10]

2.1.8 “Multiband MIMO Microwave and Millimeter Antenna System Employing Dual-function Tapered Slot Structure” - Muhammad Ikram et.al.

A single-layered multiple-input multiple-output (MIMO) antenna system is presented for future wireless devices operating at WLAN (2.45, 5.2 GHz), 4G LTE (2.6 GHz) and 5G (24, 28 GHz) bands. The design consists of a dual-band 4-element monopole MIMO antenna system that covers 2.45 and 5.2 GHz bands. A decoupling mechanism is designed in the ground plane based on a tapered slot. This tapered slot has a dual function, working both as a decoupling structure ($\lambda_0/4$ long) at 2.45 GHz and as a tapered slot antenna ($3\lambda_0$ long) at 28 GHz when excited by a proper feeder. The proposed MIMO antenna system is designed on an RO-5880 substrate with overall dimensions of $104 \times 104 \times 0.51$ mm³. The measured results show that the presented design covers two low frequency bands (2.40–2.80, 5.1–5.6 GHz) with peak gain of 5 dBi and one high frequency band (23–30 GHz) with peak gain of 11 dBi. The minimum measured isolation between the two antennas is more than 16 dB, while the maximum envelope correlation coefficient value is less than 0.16 showing good MIMO characteristics.[11]

2.1.9 “Isolation improvement of UWB MIMO antenna utilising molecule fractal structure” - S. Rajkumar et.al.

A compact four-element UWB MIMO antenna is proposed and its characteristics are investigated. The proposed antenna has a hexagon molecule-shaped fractal structure as its radiating element. The antenna elements are placed

orthogonally to each other in order to yield good isolation, without any additional decoupling structure. The band rejection in WLAN band is attained by etching a C-shaped slot on each radiating element. The antenna has a compact size of $40 \times 40 \times 1.6 \text{ mm}^3$ and exhibits stable omni-directional radiation pattern. In addition, it shows an acceptable impedance bandwidth ($S_{11} < -10 \text{ dB}$) in the range 2.4–10.6 GHz with an isolation better than -20 dB over the UWB range. The performance of the proposed antenna is calculated in terms of antenna characteristics and MIMO

parameters. The isolation level in-between the elements is found to be reasonable for MIMO application. The simulated and measured results are in good accord which makes the antenna a suitable candidate for massive MIMO and high-density packaging applications.[12]

2.1.10 “Eight-element UWB-MIMO array with three distinct isolation mechanisms” - R. Saleem et.al.

An eight-element multiple-input–multiple-output (MIMO) antenna array for ultra-wideband (UWB) communication is reported. The proposed array contains eight antenna elements having annular ring slots in the middle of each element and tapered feed lines for impedance matching. The defected ground planes are introduced on the back of these antenna elements. Moreover, closed-loop frequency selective surfaces are introduced to achieve enhanced isolation between

antenna pairs. A quad-strip connected circular arc decoupling structure is also used to enhance the isolation among particular antenna elements. These decoupling structures provide isolation of 20 dB over most of the desired UWB spectrum. The proposed UWB-MIMO array has dimensions of $60 \times 93 \text{ mm}^2$. The proposed design is suitable for integration with system-in-package applications and provides ease of connectivity on all four sides.[13]

2.1.11 “Compact 4×4 UWB-MIMO antenna with WLAN band rejected operation” – MS Khan et.al.

A compact planar UWB-MIMO antenna array with WLAN band rejection is presented. The array consists of four monopole radiators and a common ground plane. These monopoles are placed in such a way that the polarisation diversity of nearly placed radiators is exploited, resulting in high isolation. The proposed MIMO antenna array is electrically small ($50 \times 39.8 \text{ mm}^2$), printed on a low loss 1.524 mm thick Rogers TMM4 laminate with a dielectric constant of 4.5 and a loss tangent of 0.002 ($\tan \delta = 0.002$). A band-stop design was inserted on the ground plane to behave similar to a LC band-stop filter and reject the WLAN band. Simulation and measurement results satisfy the return loss requirement of better than 10 dB and isolation better than 17 dB over the entire 2.7–5.1 and 5.9–12 GHz bandwidths.[14]

2.1.12 “Compact Four-element 8-Shaped Self-Affine Fractal UWB MIMO Antenna”- Rohit Gurjar et.al.

A compact four-element 8-shaped self-affine fractal UWB MIMO antenna is presented. The self-affine fractal structure provides miniaturization and wideband characteristics to the antenna. The 8-shaped self-affine fractal geometry is designed up to the second iteration by iterated function system(IFS). This reduces the single-element antenna area by 24.88%.The dimension of proposed compact MIMO antenna is $44 \times 44 \text{ mm}^2$ and its impedance bandwidth ranges ($S_{11} < -10 \text{ dB}$) from 3.1 GHz to 11 GHz. The mutual coupling is lesser than -16 dB across the entire operating frequency band. The correlation coefficient is less than 0.1 and diversity gain is more than 9.95 across the operating frequency band. Results of this antenna indicate that it

is suitable for UWB applications.[15]

2.2 Literature Review

From the above literature surveys it is found out that most of the proposed antenna working in wideband and ultra wide band frequency range either are not compact or they have high mutual coupling. In this project we aim to propose a model that will serve both the purpose.

Chapter 3

Details

of

the Project

3.1 Basic Theory & Definitions

3.1.1 Microstrip Antenna

A microstrip antenna (also known as a printed antenna) usually means an antenna fabricated using photolithographic techniques on a printed circuit board (PCB) [16]. An individual microstrip antenna consists of a patch of metal foil of various shapes (a patch antenna) on the surface of a PCB (printed circuit board), with a metal foil ground plane on the other side of the board. Most microstrip antennas consist of multiple patches in a two-dimensional array. The antenna is usually connected to the transmitter or receiver through foil microstrip transmission lines. The radio frequency current is applied (or in receiving antennas the received signal is produced) between the antenna and ground plane. The Microstrip patch antenna has a dielectric substrate with a radiating patch and the feed lines are etched on one side and a ground plane on the other side as shown in Fig-1. The shape of the patch is not constrained (could be square, rectangular, circular, triangular or elliptical) and it is generally made of conducting material such as copper or gold.

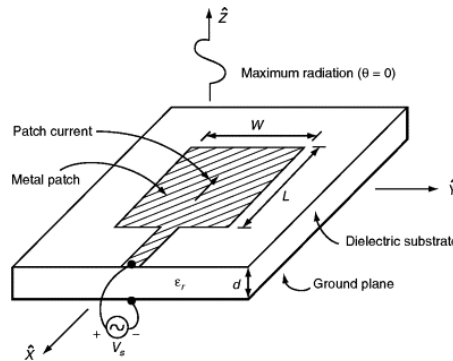


Fig-1. Structure Microstrip Patch Antenna

The fringing fields between the patch edge and the ground plane cause the microstrip patch antennas to radiate. A better performance in the antenna calls for a thick dielectric substrate having low dielectric constant which provides better efficiency, larger bandwidth and better radiation [17]. However, such a configuration results in large size of antenna. The design of a compact microstrip patch antenna demands higher dielectric constants, which are less efficient and result in narrower bandwidth. Therefore an optimization is to be achieved between antenna dimensions and antenna performance.

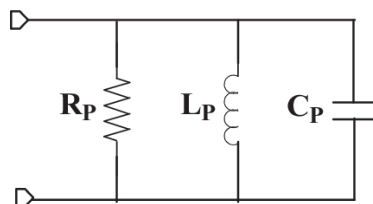


Fig-2. Equivalent Circuit Diagram of Microstrip Patch Antenna

3.1.2 Microstrip Line Feed

A microstrip line consists of a single ground plane and a thin strip conductor on a low-loss dielectric substrate above the ground plane. Due to the absence of a top ground plate and the dielectric substrate above the strips the electric lines of field remain partially in the air and partially in the lower dielectric substrate. This makes the mode of propagation called quasi-TEM. The radiation loss occurs due to the open structures and discontinuity. The use of thin

fields are mostly confined inside the dielectric. The thickness of the dielectric substrate being used increases surface waves and spurious feed radiation which hampers the bandwidth of the antenna [17]. The feed radiation also results in undesired cross polarized radiation.

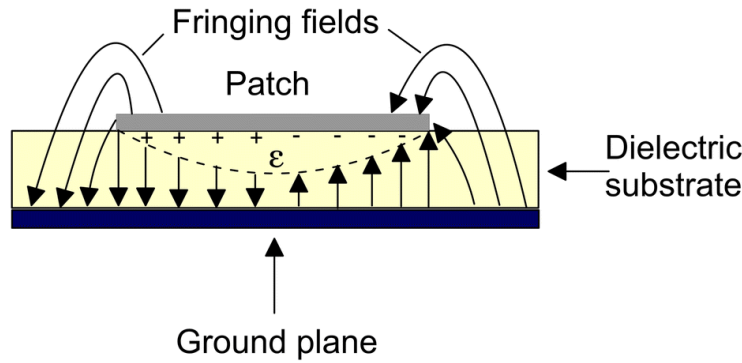


Fig-3. Fringing field lines in Microstrip Patch

Because of the fringing effect some of the waves travel in the substrate and some in the air, and there is an effective dielectric constant which account for fringing and the wave propagation is given by

$$\epsilon_{reff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left[1 + 12 \frac{h}{w} \right]^{-\frac{1}{2}} \quad ; \quad \frac{w}{h} > 1 \quad \text{-----(1)}$$

$$\epsilon_{reff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(\left[1 + 12 \frac{h}{w} \right]^{-\frac{1}{2}} + 0.04 \left[1 - \frac{w}{h} \right]^2 \right) \quad ; \quad \frac{w}{h} < 1$$

------(2)

3.1.3 Advantage and Limitations of Microstrip Antenna

Microstrip antennas have many advantages compared to conventional microwave antennas which are listed as follows:

- Light weight and low volume.
- Low profile planar configuration which can be easily made conformal to host surface.
- Low fabrication cost, hence can be manufactured in large quantities.
- Supports both, linear as well as circular polarization.
- Can be easily integrated with microwave integrated circuits (MICs).
- Capable of dual and triple frequency operations.
- Mechanically robust when mounted on rigid surfaces.

However, microstrip antennas have the following limitations when compared with microwave antennas:

- Narrow bandwidth
- Low efficiency and gain
- Extraneous radiation from feeds and junctions.
- Poor end fire radiator except tapered slot antennas.
- Low power handling capacity.
- Surface wave excitation.

3.2 Fundamental Antenna Parameters

3.2.1 Bandwidth

Bandwidth (BW) is the range of frequencies within which the performance of the antenna, with respect to some characteristic, conforms to a specified standard. The bandwidth can be considered to be the range of frequencies, on either side of the center frequency, where the antenna characteristics are within an acceptable value of those at the center frequency. Generally, in wireless communications, the antenna is required to provide a return loss less than -10dB over its frequency bandwidth. The frequency bandwidth of an antenna can be expressed as either absolute bandwidth (ABW) or fractional bandwidth (FBW). If f_H and f_L denote the upper edge and the lower edge of the antenna bandwidth, respectively. The ABW is defined as the difference of the two edges and the FBW is designated as the percentage of the frequency difference over the center frequency, as given in Equation (3) .

$$ABW = f_H - f_L \quad \text{-----}(3)$$

3.2.2 Radiation Pattern

The radiation pattern (or antenna pattern) is the representation of the radiation properties of the antenna as a function of space coordinates. In most cases, it is determined in the far-field region where the spatial (angular) distribution of the radiated power does not depend on the distance. Usually, the pattern describes the normalized field (power) values with respect to the maximum values. The radiation property of most concern is the two- or three-dimensional (2D or 3D)

spatial distribution of radiated energy as a function of the observer's position along a path or surface of constant radius. In practice, the three-dimensional pattern is sometimes required and can be constructed in a series of two-dimensional patterns. For most practical applications, a few plots of the pattern as a function of φ for some particular values of frequency, plus a few plots as a function of frequency for some particular values of θ will provide most of the useful

information needed, where φ and θ are the two axes in a spherical coordinate system. For a linearly polarized antenna, its performance is often described in terms of its principle E plane and H-plane patterns. The E-plane is defined as the plane containing the electric-field vector and the direction of maximum radiation whilst the H-plane is defined as the plane containing the magnetic-field vector and the direction of maximum radiation [18]

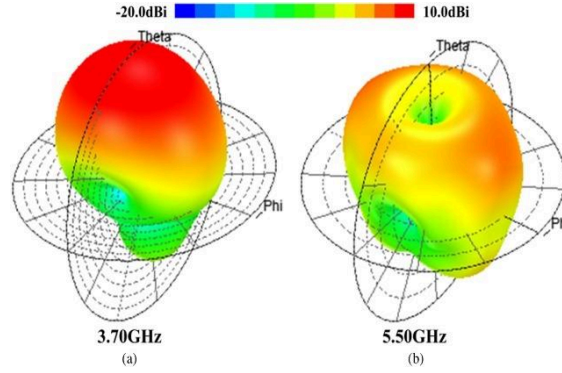


Fig-4. Radiation Pattern of Dual Band Microstrip Patch Antenna (Zijian Xing et.al.)

3.2.3 Directivity

The directivity $D(\theta, \varphi)$ is a far-field parameter of an antenna defined in a given direction (θ, φ) of an antenna. it is defined as the ratio of the radiation intensity U in a given direction from the antenna over that of an isotropic source. And it is given by

$$D(\theta, \varphi) = \frac{U(\theta, \varphi)}{U_{av}} = \frac{4\pi U(\theta, \varphi)}{P_{rad}} \quad \text{-----(4)}$$

If the direction is not specified then the Directivity is given as

$$D_0 = \frac{U_{max}}{U_{av}} = \frac{4\pi U_{max}}{P_{rad}} \quad \text{-----(5)}$$

3.2.4 Antenna Gain

The antenna absolute gain according to is defines as “the ratio of the intensity, in a given direction, to the radiation intensity that would be obtained if the power accepted by the antenna were radiated isotropically.”

$$G(\theta, \varphi) = \frac{P(\theta, \varphi)}{P_{isotropic}} = \frac{4\pi U(\theta, \varphi)}{P_{in}} \quad \text{-----(6)}$$

3.2.5 VSWR

VSWR stands for Voltage Standing Wave Ratio, and is also referred to as Standing Wave Ratio (SWR). VSWR is a function of the reflection coefficient, which describes the power reflected from the antenna. If the reflection coefficient is given by Γ , then VSWR is defined as: $VSWR = \frac{1+|\Gamma|}{1-|\Gamma|}$ -----(7)

The VSWR is always a real and positive number for antennas. The smaller the VSWR is, the better the antenna is matched to the transmission line and the more power is delivered to the antenna. The minimum VSWR is 1.0. In this case, no power is reflected from the antenna, which is ideal. Often antennas must satisfy a bandwidth requirement that is given in terms of VSWR.

3.2.6 S-Parameter

S-parameters describe the input-output relationship between ports (or terminals) in an electrical system. For instance, if we have 2 ports (intelligently called Port 1 and Port 2), then S_{12} represents the power transferred from Port 2 to Port 1. S_{21} represents the power transferred from Port 1 to Port 2. In general, S_{NM} represents the power transferred from Port M to Port N in a multi-port network.

A port can be loosely defined as any place where we can deliver voltage and current. So, if we have a communication system with two radios (radio 1 and radio 2), then the radio terminals (which deliver power to the two antennas) would be the two ports. S_{11} then would be the reflected power radio 1 is trying to deliver to antenna 1. S_{22} would be the reflected power radio 2 is attempting to deliver to antenna 2. And S_{12} is the power from radio 2 that is delivered through antenna 1 to radio 1. Note that in general S-parameters are a function of frequency (i.e. vary with frequency).

3.2.7 Envelope Correlation Coefficient(ECC)

Envelope Correlation Coefficient tells us how independent two antennas' radiation patterns are. So if one antenna was completely horizontally polarized, and the other was completely vertically polarized, the two antennas would have a correlation of zero. Similarly, if one antenna only radiated energy towards the sky, and the other only radiated energy towards the ground, these antennas would also have an ECC of 0. Hence, Envelope Correlation Coefficient takes into account the antennas' radiation pattern shape, polarization, and even the relative phase of the fields between the two antennas. ECC is calculated using the expression

$$\rho_{ij} = \frac{|S_{ii}^* S_{ij} + S_{ji}^* S_{jj}|^2}{[(1 - (|S_{ii}|^2 + |S_{ji}|^2))(1 - (|S_{jj}|^2 + |S_{ij}|^2))]} \quad \text{-----(8)}$$

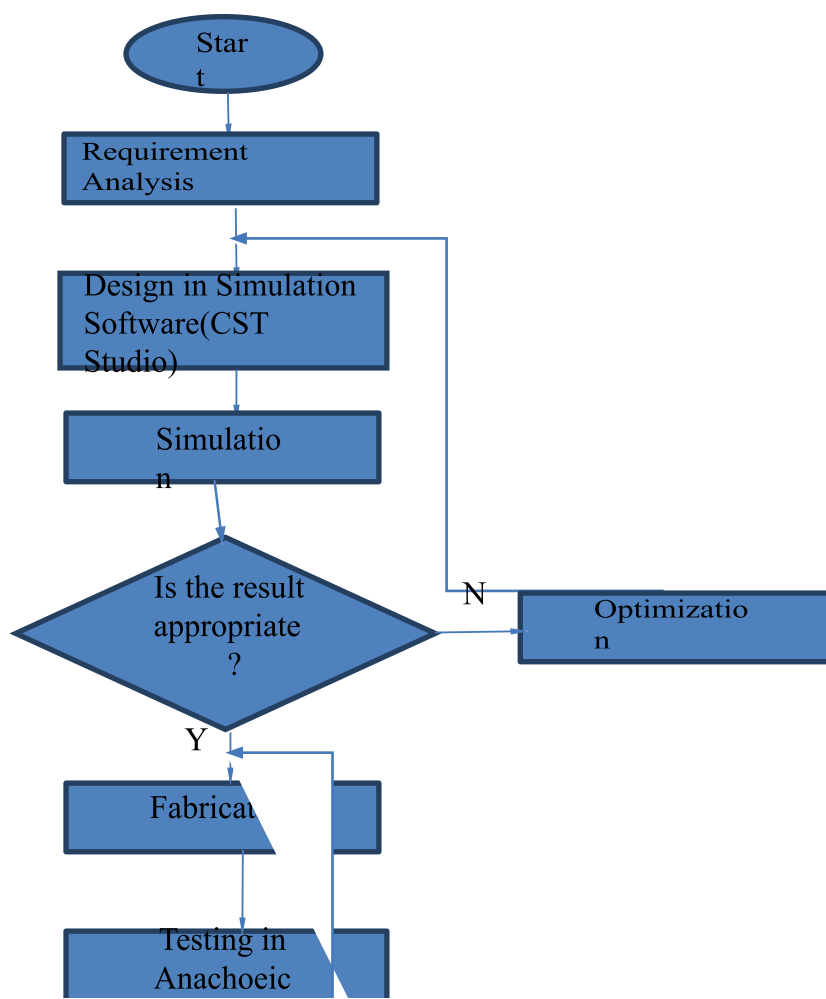
3.2.8 Diversity Gain

Diversity gain is the increase in signal-to-interference ratio due to some diversity scheme, or how much the transmission power can be reduced when a diversity scheme is introduced, without a performance loss. Diversity gain

is calculated using the expression $DG = 10\sqrt{1 - ECC^2}$ -----(9)

The acceptable value of diversity gain should be close to 10dB.

3.3 Flow Chart



3.4 Antenna Design

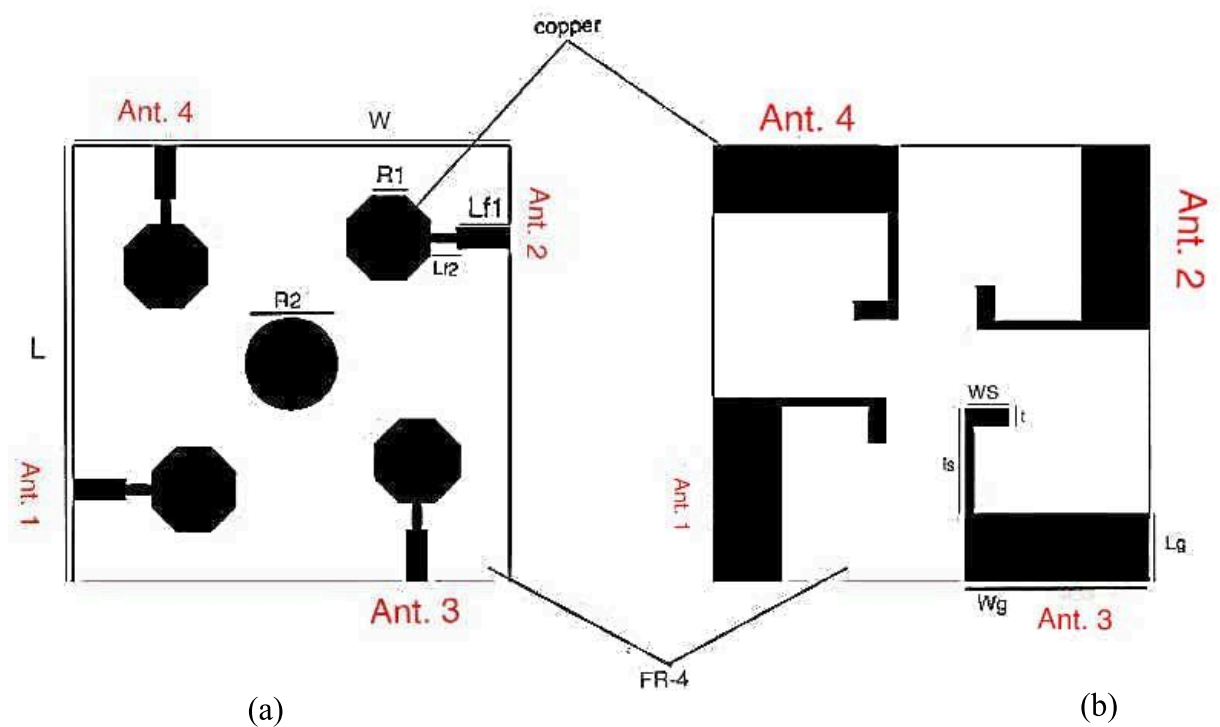


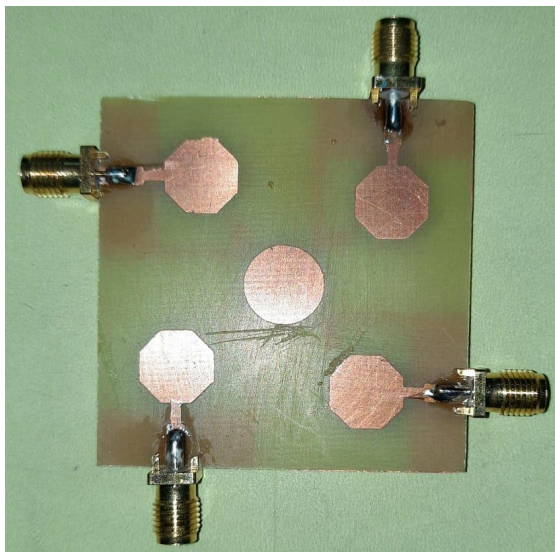
Fig 5. (a) Top View and (b) Bottom View of proposed MIMO Antenna

3.4 Antenna Design Parameters

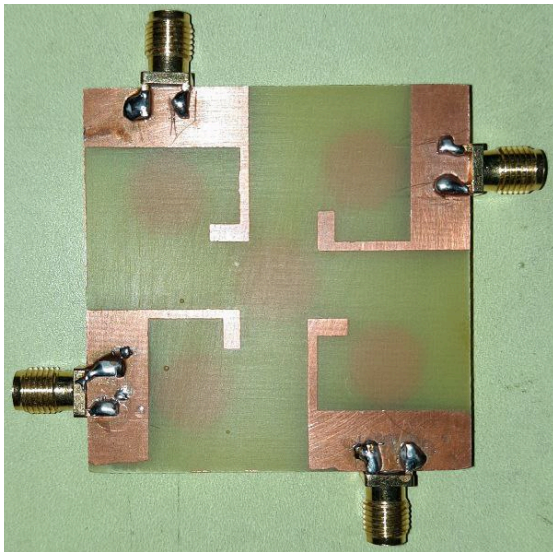
Sl. No.	Parameters	Dimension (in mm)
1	R_1	4
2	R_2	10.67
3	L_{f1}	6
4	L_{f2}	3
5	L_g	7.8
6	W_g	21
7	w_s	3
8	W_{f1}	2.3
9	W_{f2}	1.15
10	L	50
11	W	50

Table 1. Antenna Parameters of the proposed MIMO Antenna

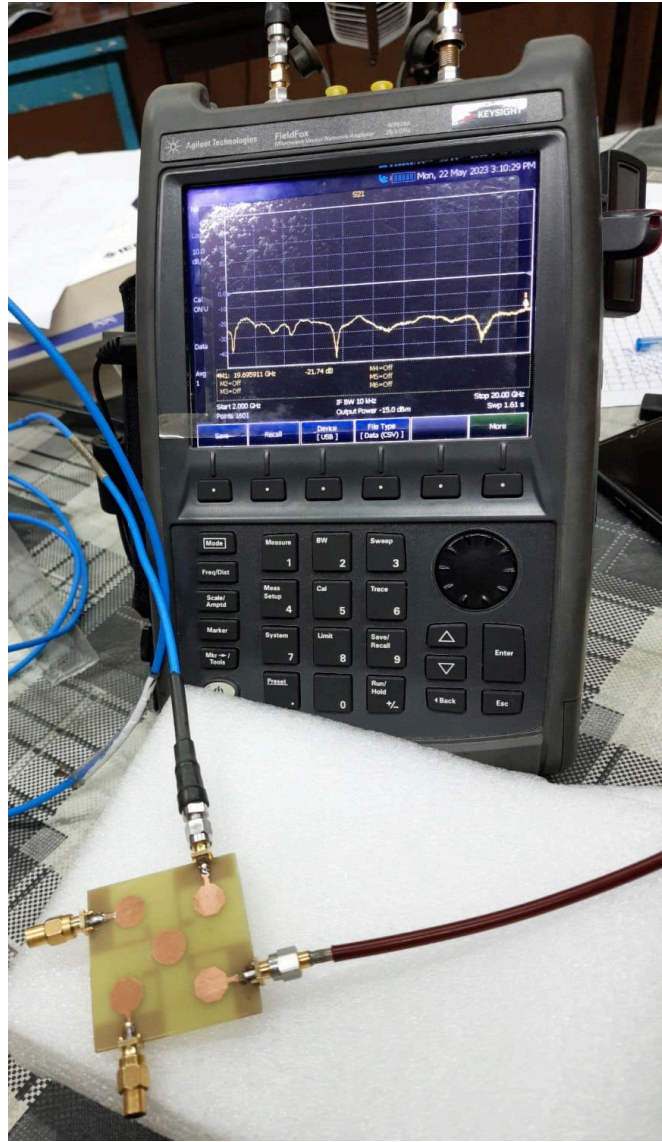
3.5 Fabricated Antenna Prototype



(a)



(b)



(c)

Fig 6. (a) Top View and (b) Bottom View of the fabricated prototype of the proposed antenna , (c) Antenna parameters measurement setup

Chapter 4

Results

4.1 Simulated Results

4.1.1 S-Parameters

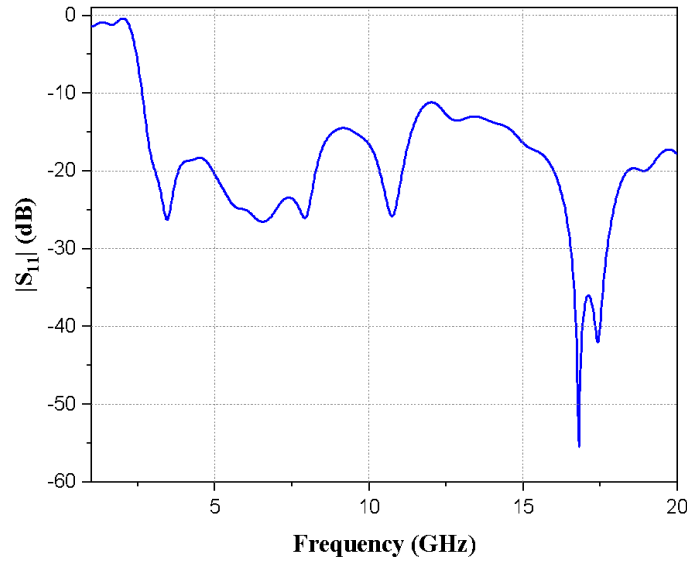


Fig-7. Simulated S_{11} of the Ant 1

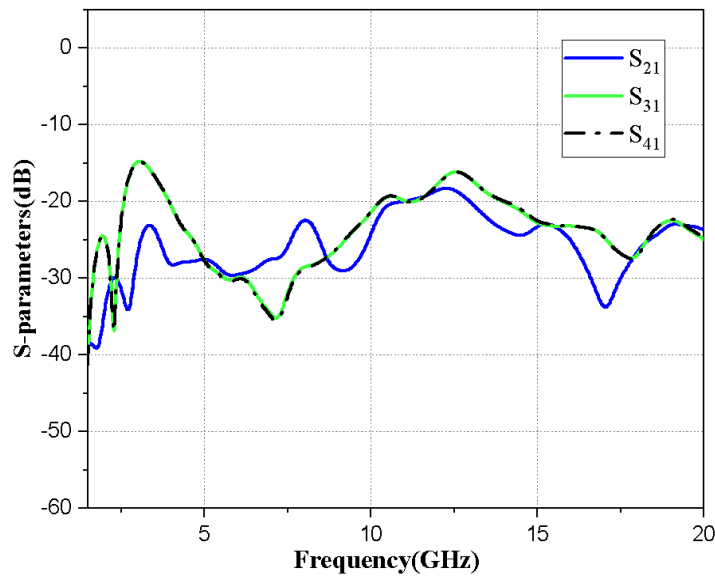


Fig-8. Simulated isolation parameters between Ant 1 and Ant 2,Ant3,Ant 4

Impedance bandwidth and isolation performance of the proposed MIMO antenna design can be expressed in terms of different S-parameters. S-parameters describe the input–output relationship between ports (or terminals) in an electrical system. Therefore, in this four element MIMO antenna system, S_{11} parameter specifies the amount of power that is reflected back to Ant. 1 element when port 1 is excited. In order to obtain less amount of reflection and better impedance matching, S_{11} should be very small. The impedance bandwidth of an antenna element is considered as the frequency range over which the value of $S_{11} \leq -10$ dB, and which gives us the impedance bandwidth from 2.64 GHz to 20 GHz . Similarly, S_{ij} parameter represents the power received at port i relative to the power input at port j . Isolation performance of a MIMO antenna configuration can be predicted from S_{ij} plots. In case of MIMO antenna systems, the amount of isolation between antenna elements should be very high which indicates very less amount of power is coupled from one port to another. Although no specific value of isolation has been defined for a good MIMO antenna

system, for most of the wide band MIMO antennas, isolation has been observed to be more than around 20 dB over the sub-6 GHz frequency range of operation. As the proposed MIMO antennacnfiguration is completely symmetric, the reflection coefficient plot for all these four antenna elements are same. Therefore, only S_{11} response is exhibited

here to describe the impedance bandwidth of the proposed design. Similarly, to represent the isolation between antenna elements, S21, S31 and S41 plots are sufficient because all other isolation curves will be identical to these three curves.

4.1.2 Radiation Patterns

4.1.2.1 At f=4.2 GHz

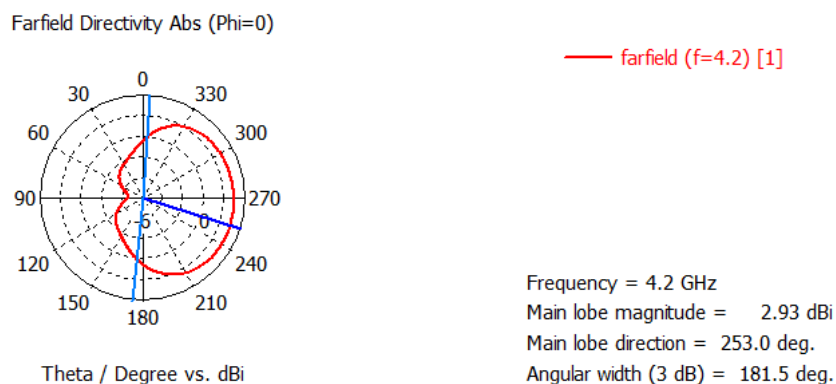


Fig-18. E-plane radiation pattern at f=4.2 GHz

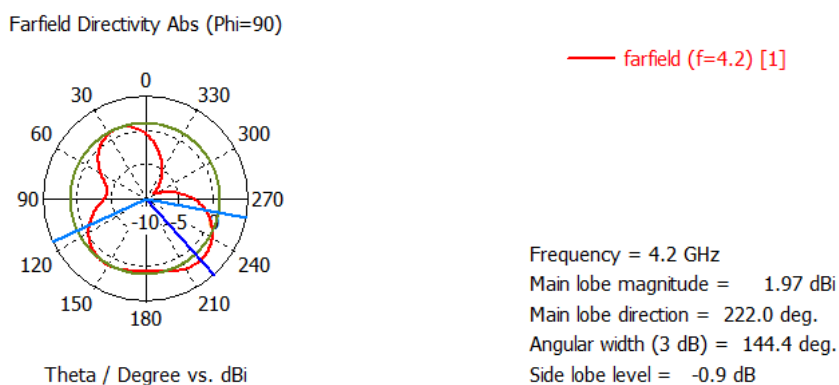
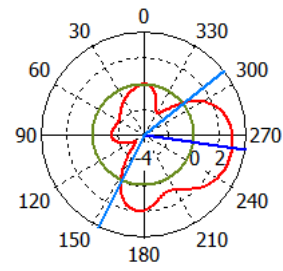


Fig-19. H-plane radiation pattern at f=4.2 GHz

4.1.52.2 At f=6 GHz

Farfield Directivity Abs (Phi=0)



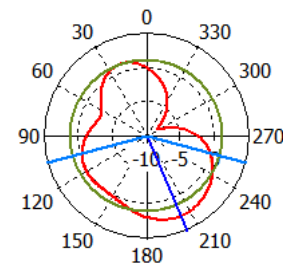
Theta / Degree vs. dBi

— farfield (f=6) [1]

Frequency = 6 GHz
Main lobe magnitude = 2.98 dBi
Main lobe direction = 262.0 deg.
Angular width (3 dB) = 154.6 deg.
Side lobe level = -3.0 dB

Fig-22. E-plane radiation pattern at f=6 GHz

Farfield Directivity Abs (Phi=90)



Theta / Degree vs. dBi

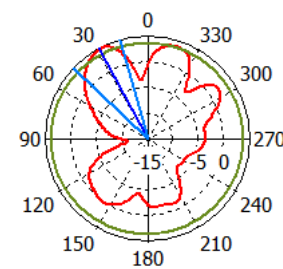
— farfield (f=6) [1]

Frequency = 6 GHz
Main lobe magnitude = 2.66 dBi
Main lobe direction = 203.0 deg.
Angular width (3 dB) = 150.6 deg.
Side lobe level = -1.5 dB

Fig-23. H-plane radiation pattern at f=6 GHz

4.1.2.3 At f=16.1 GHz

Farfield Directivity Abs (Phi=0)



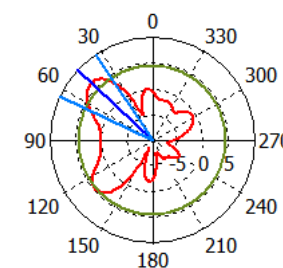
Theta / Degree vs. dBi

— farfield (f=16.1) [1]

Frequency = 16.1 GHz
Main lobe magnitude = 4.81 dBi
Main lobe direction = 28.0 deg.
Angular width (3 dB) = 30.8 deg.
Side lobe level = -1.0 dB

Fig-24. E-plane radiation pattern at f=16.1 GHz

Farfield Directivity Abs (Phi=90)



Theta / Degree vs. dBi

— farfield (f=16.1) [1]

Frequency = 16.1 GHz
Main lobe magnitude = 6.21 dBi
Main lobe direction = 47.0 deg.
Angular width (3 dB) = 31.1 deg.
Side lobe level = -1.7 dB

Fig-25. H-plane radiation pattern at f=16.1 GHz

4.1.3 Maximum Gain Over Frequency

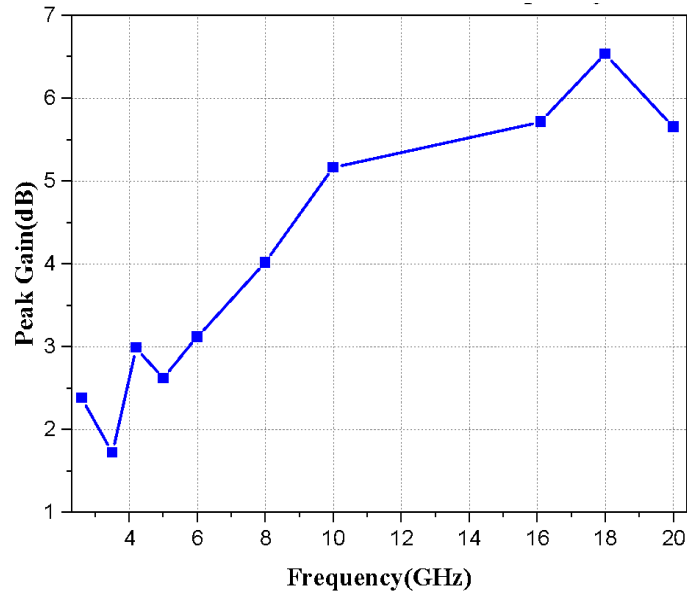


Fig-12. Maximum Gain Over Frequency

The peak gain achieves 6.53 dB at 18 GHz frequency.

4.1.4 Diversity Gain

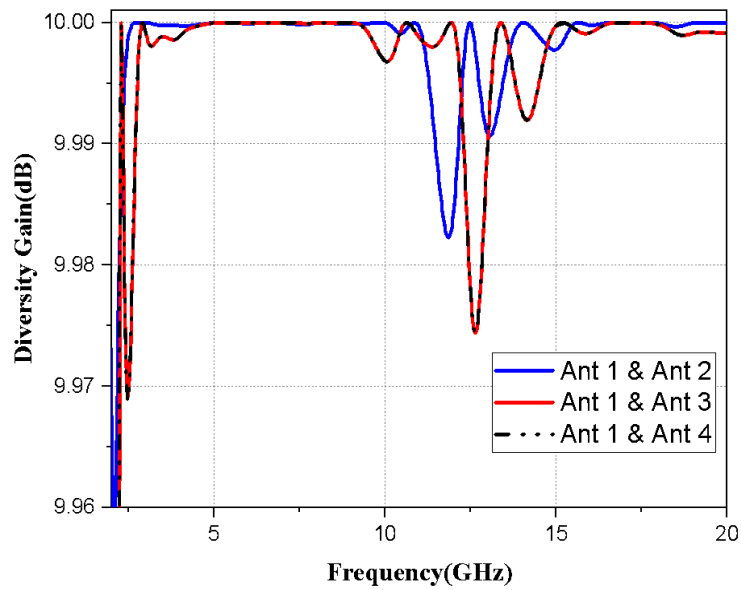


Fig-13. Diversity Gain between Antennas

Diversity Gain achieved for the MIMO antenna, which illustrates the reduced amount in transmission power after using a diversity scheme without a performance loss for the MIMO systems. Moreover, it defines the amount of improvement gained by using a multiple antenna system compared to a single antenna system. The values are nearly 10 dB (ideal value) across the operating frequency bands and in moreover sub-6 GHz 5G band.

4.1.5 Envelope Co-relation Co-efficient (ECC)

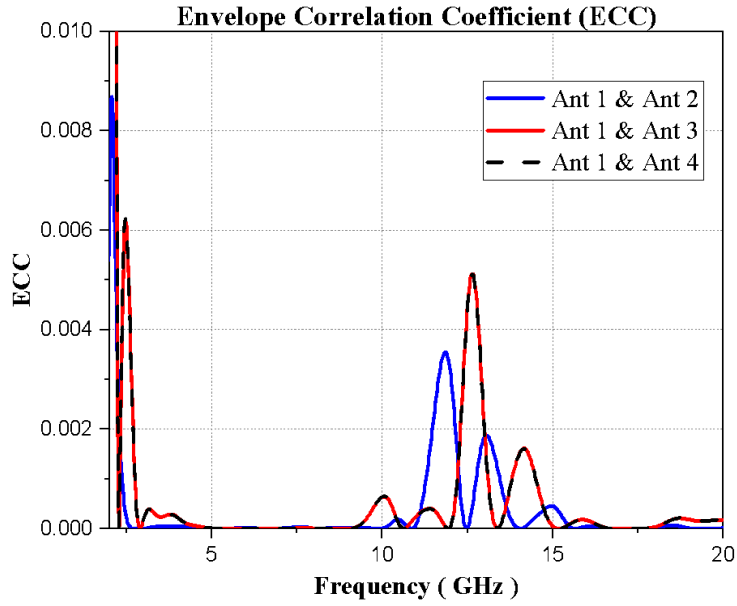


Fig-14. ECC between Antennas

Fig-14 shows the ECC curve of the proposed antenna over frequency, relatively larger values of ECC are shown between antennas 1 and 2 as well as 3 and 4. The overall antenna module ensures that there are correlation values below the practical standard of 0.5. The ECC values of the two antenna elements are always below 0.01 over the sub-6 GHz 5G frequency band, implying acceptable diversity performance.

4.2 Measured Results

4.2.1 Reflection Parameter

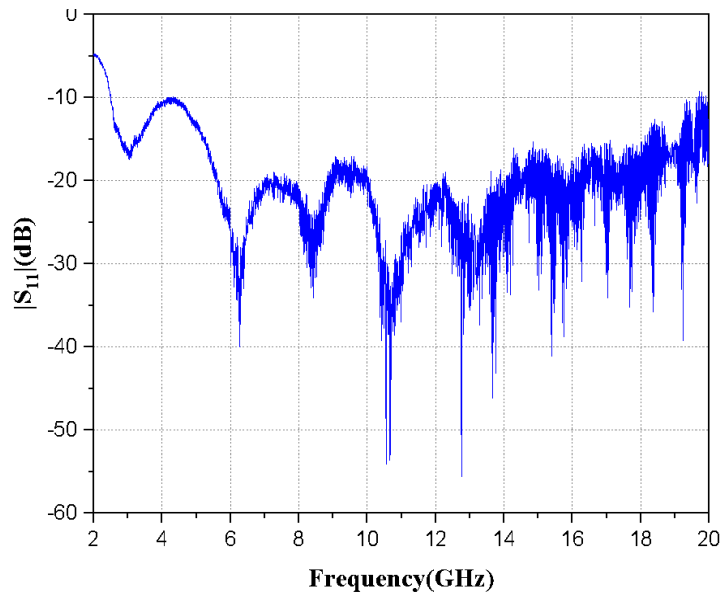


Fig-15. Measured S_{11} of the Ant 1

The measured impedance bandwidth is achieved from 2.51 GHz to 19.84 GHz

4.2.2 Isolation Parameter

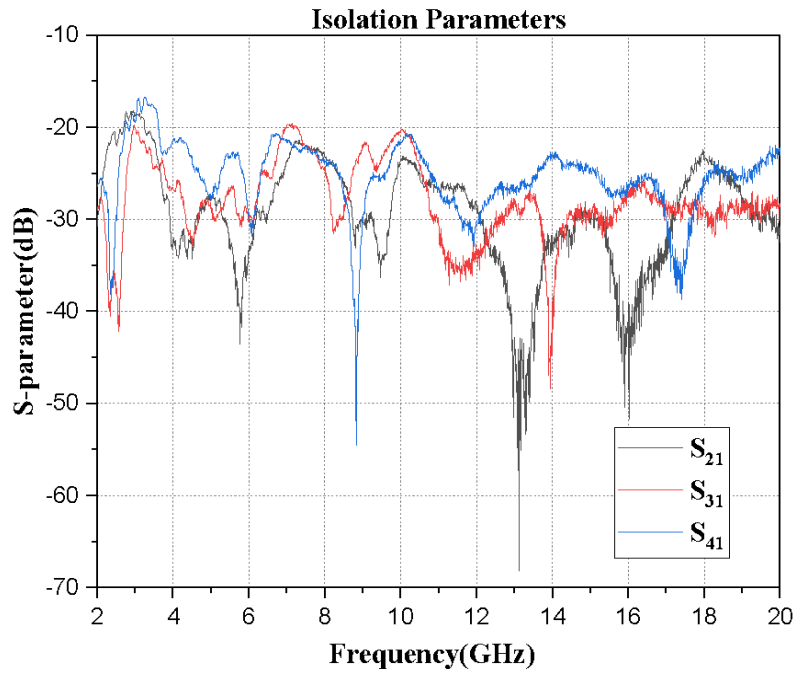


Fig-16. Measured isolation parameters between Ant 1 and Ant 2,Ant3,Ant 4

The measured isolation achieved is greater than 20dB in sub-6 GHz bands

4.3 Comparison between Simulated and Measured Results

4.3.1 Reflection Parameter

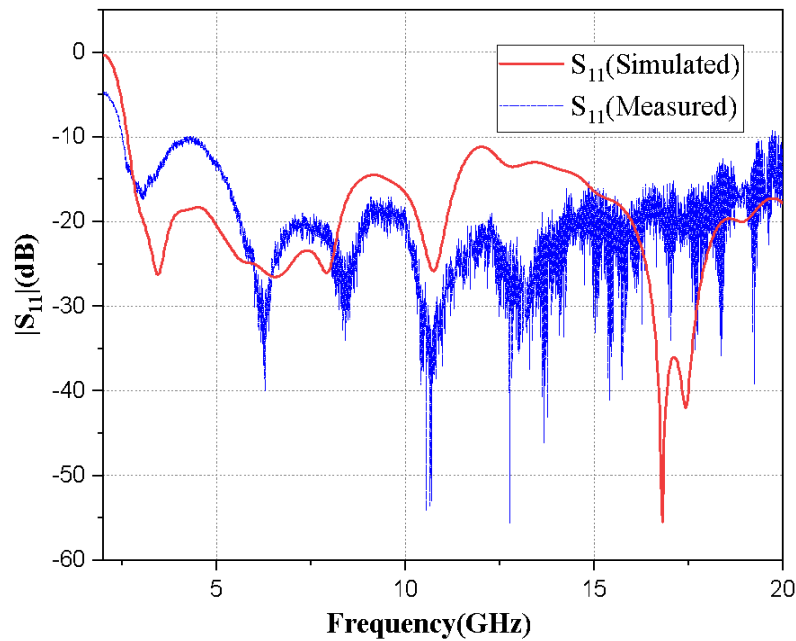


Fig-17. Comparison between Simulated and Measured S_{11} of the Ant 1

4.3.2 Isolation Parameter

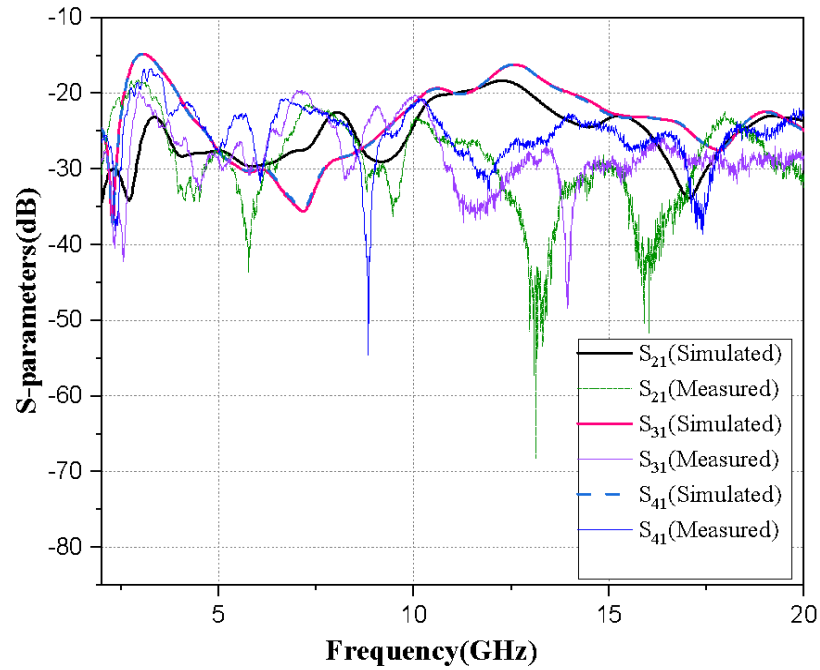


Fig-17. Comparison between Simulated and Measured isolation parameters between Ant 1 and Ant 2, Ant 3, Ant 4

The difference between the simulated and measured results is due to fabrication imperfections and tolerance due to the permittivity of the substrate and port/cable coupling losses.

Chapter 5

Comments on Results

A compact printed triple band-notched Wide Band(WB) MIMO antenna with high isolation and enhanced bandwidth has been presented and analyzed for portable WB applications. The proposed antenna has an operational impedance bandwidth for $|S_{11}| \leq -10$ dB ranging from 2.6 GHz to 20 GHz ($VSWR \leq 2$). The simulated and measured results show good resemblance and have shown that the antenna has high isolation of >20 dB over the entire sub-6 GHz. The focus of this design is to develop a compact printed WB MIMO antenna covering the entire IoT and sub-6 GHz bands improving isolation between the antenna elements. The proposed antenna shows good diversity performance as the $ECC \leq 0.001$ for the entire wide bandwidth. The computed and measured gain variations are stable over the entire WB range, except at the notch frequencies, and achieves a peak gain of 6.53 dB at 18 GHz. All the simulated and measured results depict that the proposed four-port compact Wide Ban MIMO antenna is a good candidate for portable wide band system for IoT and 5G application.

Chapter 6

Conclusion

6.1 Conclusions

In this project, a compact four-element, four-port MIMO antenna system is demonstrated for IoT and 5G applications. The structure of the presented antenna, S-parameters, parametric studies, radiation patterns, and the effect of the MIMO configurations are thoroughly discussed using a simulated design and fabricated prototype. According to simulated and measured results, the design exhibited an isolation of more than 20 dB at the sub-6GHz 5G frequency bands, while it has a compact structure. Additionally, the proposed design has a high peak gain of 6.53 dBi is achieved at 18 GHz frequency. In addition, an ECC less than 0.01 and other MIMO performance metrics showed that the proposed antenna is suitable for diversity performance. Therefore, the above characteristics validate that the proposed antenna is a good candidate for multiband MIMO applications covering 2.64 GHz–20 GHz bands.

6.2 Future Works

- The Bandwidth of the antenna will be enhanced by using several techniques.
- The Isolation of the antenna will be enhanced by different approach.
- The Gain of the antenna will be enhanced.
- Efficiency of the antenna will be enhanced.
- The optimization of the antenna will be done by different Machine Learning algorithms.

6.3 Gantt Chart of the Project

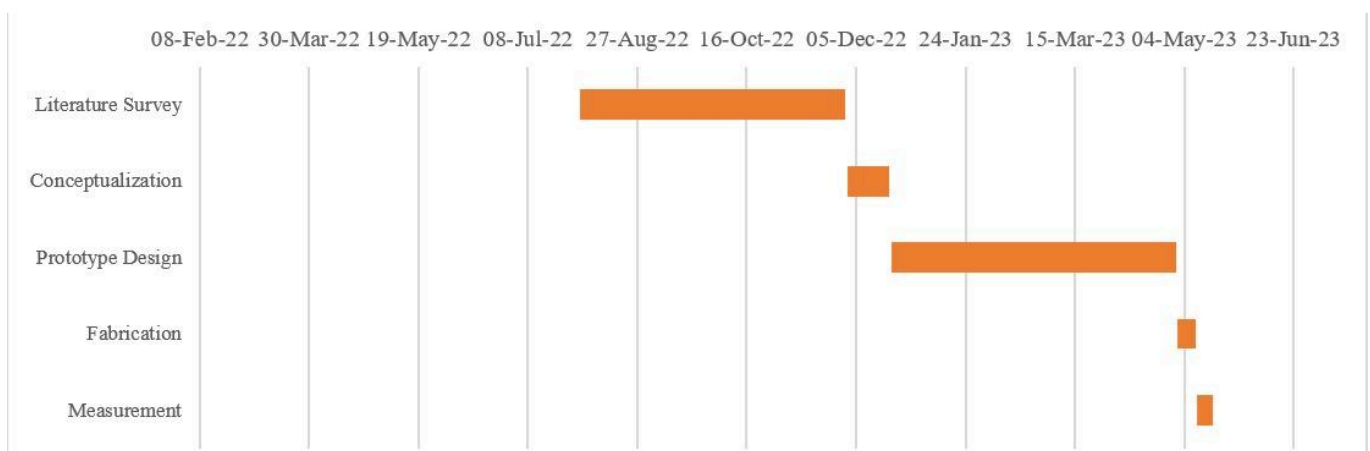


Fig-18. Gantt Chart of the Project

6.4 Expenses and Financing

Sl. No	Item	Quantity	Cost
1.	Female SMA Connector	10	850

2.	Antenna (Fabrication cost)	1	2000
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Chapter 7

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